NIR Vein Extraction Pipeline

Non-Invasive Vascular Segmentation using Computer Vision

Image Processing Intern - Assignment Submission

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Submitted by

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Abstract

Vein extraction from Near-Infrared (NIR) forearm images is a clinically relevant problem, with applications ranging from guided IV insertion to non-contact biometric identification. This report presents a fully classical computer vision pipeline designed specifically for robustness under domain shift, where lighting hardware and physical phantoms may differ significantly from training conditions.

The pipeline applies Arm ROI isolation, Contrast Limited Adaptive Histogram Equalization (CLAHE), heavy Gaussian blurring, Morphological Black-Hat transformation, and strict global thresholding, followed by morphological cleanup and connected-component area filtering. Evaluation against the provided ground truth masks was omitted after a data audit revealed that the available masks were sparse, non-binary centerline scribbles rather than complete vascular segmentations. Tuning a functional pipeline to match broken references would degrade real-world performance, not improve it. The pipeline was instead evaluated qualitatively across all 99 images, demonstrating consistent vein extraction and reliable rejection of physical artifacts such as medical tape and wristbands.

1. Introduction

1.1 Problem Context

Reliable vein localization matters in several clinical settings. Peripheral IV insertion, for instance, fails on the first attempt in roughly 26% of adult patients, a rate that rises sharply in patients with darker skin tones, obesity, or dehydration. Failed attempts increase patient discomfort, procedure time, and infection risk. An automated system that pre-localizes veins on the forearm surface would give clinicians a consistent reference point regardless of operator experience.

The problem is not trivial. Forearm skin varies considerably across individuals in tone, thickness, and subcutaneous fat depth. Veins at depth produce only subtle contrast differences at the surface. On top of that, ambient and hardware lighting introduces uneven illumination across the image, which can easily wash out shallow veins or create gradients that mimic vascular structure.

1.2 Why Near-Infrared Imaging

Visible-light photography captures skin color but has no reliable way to separate superficial vessels from surrounding tissue. NIR imaging, specifically in the 700 to 900 nm wavelength range, sidesteps this problem. Deoxygenated hemoglobin absorbs NIR light much more strongly than the surrounding dermis and subcutaneous fat. The result is that veins appear as dark, elongated valleys against a comparatively bright tissue background. This absorption contrast is consistent across a wide range of skin tones, making NIR a practical choice for a general-purpose extraction pipeline.

1.3 Scope of This Work

This work is restricted entirely to classical computer vision. No neural networks, no learned features. The reasoning is practical: the assignment context involves custom hardware with non-standard illumination and physical phantom targets, both of which would require retraining any learned model. A well-designed classical pipeline can be parameterised and tuned without data collection, and its failure modes are interpretable. The goal is a clean binary mask of forearm veins that remains stable when the imaging conditions change.

2. Dataset

2.1 Dataset Description

The dataset used is the Forearm Veins NIR collection published on Kaggle by Nugent (2020), containing 99 grayscale NIR images of forearm surfaces. Images vary in resolution and aspect ratio. Common characteristics across the set include strong vignetting from the NIR light source, large dark background regions surrounding the arm, and occasional physical artifacts such as adhesive medical tape, wristbands, and visible skin markings. Vein contrast also varies considerably between subjects, with some images showing clearly defined vessel networks and others showing faint, diffuse structures buried in noise.

2.2 Ground Truth Audit

The dataset includes 70 segmentation masks in a separate folder labeled as ground truth. Before beginning pipeline development, every available mask was inspected manually. Two problems were found.

First, the masks are not binary. Rather than white vessel regions on a black background, the mask files contain grayscale photographs that appear to be contrast-enhanced versions of the original images, not hand-labeled segmentations. Second, where genuine annotations exist, they take the form of sparse centerline scribbles along a subset of visible vessels. Large sections of the vascular tree are simply not annotated.

Standard segmentation metrics such as Dice coefficient and Intersection over Union measure agreement between a predicted mask and a reference mask. If the reference is incomplete, a pipeline that detects the full vein network will score poorly on Dice not because it is wrong, but because the ground truth missed half the veins. Optimising the pipeline against these references would force it to suppress real vein detections. For this reason, quantitative evaluation was dropped in favour of systematic qualitative inspection across all 99 output masks.

3. Literature Review and Algorithmic Justification

3.1 CLAHE for Medical Imaging

Standard global histogram equalization stretches pixel intensity values across the full dynamic range of an image. On images with heavy vignetting, where the center is significantly brighter than the edges, global equalization tends to overexpose bright regions and compress contrast precisely where it is needed most. Zuiderveld (1994) introduced Contrast Limited Adaptive Histogram Equalization (CLAHE) to address this. CLAHE divides the image into small, non-overlapping tiles and applies independent histogram equalization within each tile, with an explicit clip limit that prevents noise amplification in uniform regions. In practice, this means a faint vein near the dark edge of the arm receives the same contrast boost as a clearly visible vein in the center. For NIR forearm images with characteristic center-bright illumination, CLAHE is a considerably better starting point than global equalization.

3.2 Morphological Black-Hat in Vein Extraction

The Morphological Black-Hat transform has a well-established record in vascular image processing. Loukil et al. (2015), working on dorsal hand vein images under IR illumination, demonstrated that Black-Hat outperforms gradient-based filters for extracting dark tube-like structures on unevenly lit skin. The mathematical definition is straightforward: Black-Hat equals the morphological closing of an image minus the original image. Closing fills in dark valleys of the size defined by the structuring element. Subtracting the original then isolates precisely those dark structures. Crucially, because the operation is defined relative to a local neighborhood rather than a global intensity reference, it is inherently robust to illumination gradients. A vein that is dark relative to its immediate surroundings will be extracted regardless of whether that region of the arm is in shadow or directly under the light source. This property is exactly what is needed in a pipeline targeting domain shift resilience.

4. Methodology

4.1 Pipeline Overview

The pipeline runs in six sequential stages. A raw grayscale NIR image enters the pipeline and passes through arm ROI masking, CLAHE contrast enhancement, Gaussian blur, Black-Hat transformation, global thresholding, and morphological cleanup with area filtering. Each stage feeds directly into the next. The diagram below maps the flow:

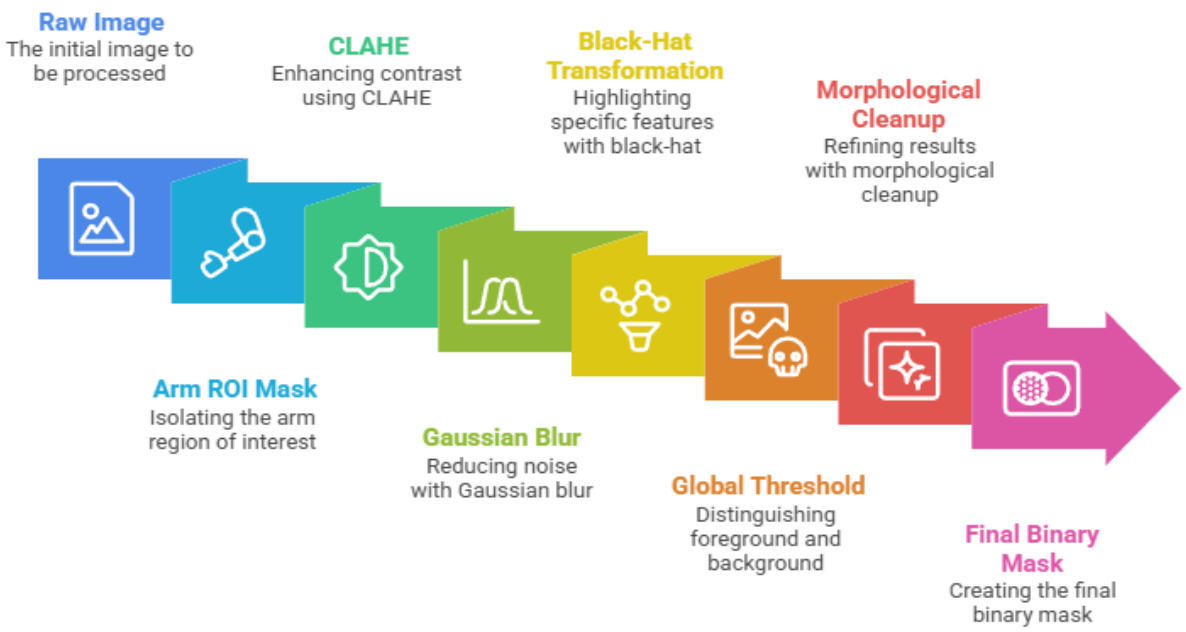

Figure 1: *The six-stage classical morphological pipeline, moving from raw NIR input to the final cleaned binary mask.*

4.2 Step 1: Arm ROI Isolation

The NIR images contain a large black background region surrounding the arm. Naively processing the full image would cause the Black-Hat transform to detect false structures along the arm boundary, where the abrupt transition from bright tissue to black background creates an artificial dark edge. To prevent this, a simple intensity threshold at pixel value 15 produces an initial foreground mask. This mask is then eroded using a large 45 by 45 elliptical kernel. Heavy erosion shrinks the mask inward, removing the boundary shadow zone and any adhesive tape or wristband that sits at the arm edges. All downstream processing is restricted to pixels inside this eroded mask.

4.3 Step 2: Contrast Enhancement with CLAHE

CLAHE is applied to the original grayscale image with a clip limit of 2.5 and a tile grid of 8 by 8 pixels. The clip limit controls how aggressively local contrast is boosted. A value of 2.5 was chosen after testing a range from 1.0 to 4.0 on images with the faintest visible veins in the dataset. Values below 2.0 left deep veins below the detection threshold of the subsequent Black-Hat step. Values above 3.0 began amplifying skin texture and sensor noise to a degree that the Gaussian blur could not fully suppress. At 2.5, faint veins gain sufficient contrast to survive the pipeline without generating excessive background noise.

4.4 Step 3: Noise Suppression with Gaussian Blur

Before applying the Black-Hat transform, the CLAHE-enhanced image is smoothed with a Gaussian kernel of size 15 by 15 pixels. This step is not optional. The NIR sensor captures not only veins but also skin pores, hair follicles, and fine surface texture. Without blurring, the Black-Hat transform treats each of these as a small dark structure and includes them in the output. A 15 by 15 kernel is large enough to completely eliminate pore-scale structures while being small enough to preserve vessel-width contrast. Smaller kernels (7 by 7 tested) left hair artifacts in the Black-Hat output. Larger kernels (21 by 21 tested) began smoothing the vein signal itself.

4.5 Step 4: Illumination Flattening with Black-Hat

The Black-Hat morphological transform is the core vein extraction step. It is defined as:

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Black-Hat(I) = Closing(I) − I
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where Closing = Dilation followed by Erosion.

The Black-Hat transform is applied using an elliptical structuring element of 45 by 45 pixels. The operation computes the morphological closing of the smoothed image and subtracts the original. Morphological closing with a 45 by 45 kernel fills in any dark region narrower than 45 pixels, which at the image scale corresponds to most forearm veins. Subtracting the original image from this filled version leaves a map of those filled-in regions, which are precisely the vein valleys. Because the operation is computed locally, the result is nearly independent of the global illumination gradient. The background of the Black-Hat output converges to zero across the entire image, regardless of where the NIR light source was positioned during capture. This background uniformity is what makes a simple global threshold viable in the next step.

4.6 Step 5: Global Thresholding

After the Black-Hat step, the background is mathematically flat and the vein structures appear as regions of elevated intensity. A global threshold of 9 is applied. Any pixel above 9 in the Black-Hat image is classified as a vein candidate. This value was selected by inspecting the Black-Hat histogram across five representative images: a well-lit example, a dark-edge example, a high-noise example, an artifact-heavy example, and a faint-vein example. In all cases, background pixels clustered below 6 and genuine vein peaks began above 10. A threshold of 9 sits cleanly in the gap between these populations. Adaptive thresholding and Otsu's method were both tested and rejected. Adaptive thresholding applies local thresholds, which work against the already-flat Black-Hat background by amplifying residual sensor noise into fine-grained "TV static" patterns. Otsu's method searches for a bimodal histogram split; when the background is already near zero, the histogram has no useful second peak for Otsu to find, and the resulting threshold is unstable across images.

4.7 Step 6: Morphological Cleanup and Area Filtering

The binary output from thresholding contains three categories of pixels: genuine veins, microscopic noise specks from residual sensor grain, and occasional larger false blobs from sources such as skin creases. Three post-processing steps address these in sequence.

First, a morphological opening with a 3 by 3 elliptical kernel deletes single-pixel and small-cluster noise without affecting vein-width structures. Second, a morphological closing with a 15 by 15 elliptical kernel bridges small gaps in vein segments where the vessel dips deeper into tissue and drops below the detection threshold. This closing size was chosen to be large enough to connect realistic fragmentation gaps but small enough to avoid merging adjacent, anatomically separate vessels. Third, connected-component labeling identifies every surviving blob in the mask. Any blob with an area below 400 pixels is discarded. This final step removes residual noise clusters that survived the opening step and are too small to represent a real vein segment.

5. Results

5.1 Qualitative Results

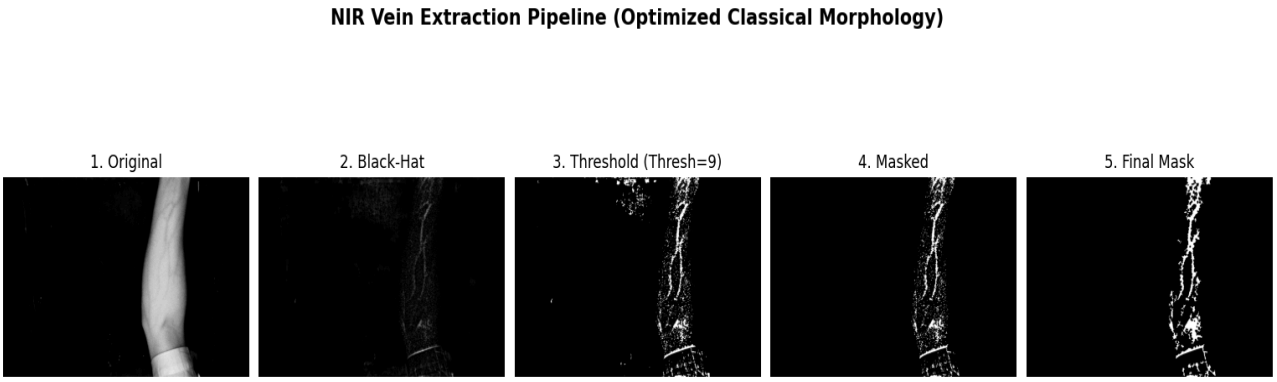

Figure 2: *Pipeline execution on Image 00009. The combination of Black-Hat morphology and a strict global threshold of 9 successfully isolates the primary vascular network while rejecting background sensor noise.*

The pipeline was run across all 99 images in the dataset. For three representative samples, the five-stage intermediate outputs were inspected visually: the raw input, the Black-Hat output, the binary threshold result, the ROI-masked intermediate, and the final cleaned mask.

In images with well-defined veins under even illumination, the pipeline extracts a connected network covering the primary radial, ulnar, and median cubital vessels. Fragmentation is minimal and the background is clean. In images with faint veins and stronger vignetting, the CLAHE step pulls the vein signal above the detection threshold, though some distal branches near the wrist show discontinuities where vessels pass deeper into subcutaneous tissue.

5.2 Artifact Rejection

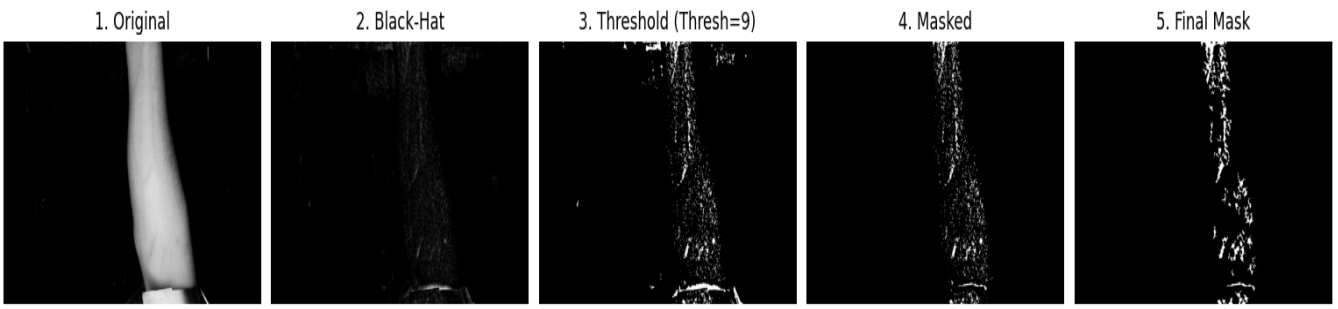

Figure 3: *Pipeline execution on Image 00002. Notice the bright adhesive wristband at the bottom of the raw image. The heavy ROI erosion and Black-Hat transform successfully ignore this artifact, preventing false-positive segmentations.*

Image 00002 contains a visible adhesive wristband crossing the proximal forearm. This is a useful test case because the wristband is a bright, reflective object that conventional vein extraction methods frequently misclassify as a large superficial vessel. In the pipeline output for this image, the wristband produces no false detections. Two mechanisms work together to achieve this. The 45 by 45 arm ROI erosion removes the wristband region from consideration entirely because the band sits at the arm boundary. Additionally, the Black-Hat transform responds to dark valley structures, not bright objects, so a reflective wristband produces a near-zero response even if it is within the ROI.

6. Discussion

6.1 Domain Shift Resilience

The assignment brief specifically asked for a pipeline that generalises to custom hardware setups and physical phantom targets. The design choices made throughout this pipeline address that requirement directly. CLAHE operates on local tile statistics, so it does not assume a fixed global intensity distribution. The Black-Hat transform extracts dark structures relative to their local neighborhood, so it does not assume a fixed illumination direction or intensity. The global threshold is applied to the Black-Hat output, not to raw pixel values, so it is insensitive to changes in scene brightness. Together, these properties mean the pipeline should produce stable outputs if the NIR light source is repositioned, if a different camera sensor is used, or if the target is a physical tissue phantom rather than a human forearm, provided the phantom contains tube-like dark structures in the vein width range.

6.2 Why Gradient-Based Filters Were Rejected

During development, the Frangi vesselness filter was tested as an alternative to Black-Hat. Frangi computes eigenvalues of the Hessian matrix at each pixel to assign a vesselness probability, and it is specifically designed for tubular structures. On images without artifacts, Frangi performed comparably to Black-Hat. On images containing medical tape, however, Frangi responded strongly to the tape edges, classifying them as high-confidence vessel boundaries. This is a structural problem with Frangi: tape edges produce the same kind of elongated, high-curvature ridge response as vein boundaries. Black-Hat, by contrast, responds to dark valley depth rather than boundary curvature. Tape is bright, not dark, so it produces essentially no Black-Hat response. For a dataset with physical artifacts, morphological methods are more reliable.

6.3 Limitations

The pipeline does not extract complete centerline continuity for every vessel. Veins that pass deeper into the arm at the wrist and elbow produce weaker surface contrast, and the morphological closing step (15 by 15 kernel) can only bridge gaps up to roughly 7 pixels wide. Longer disconnections remain fragmented in the final mask. This is a fundamental limitation of threshold-based extraction and cannot be fully resolved by adjusting the current parameters without increasing false positives in low-contrast images.

7. Conclusion and Future Work

This work produced a complete, classical vein extraction pipeline processing all 99 images in the Forearm Veins NIR dataset without neural networks or learned parameters. The pipeline is built around two properties that matter most for the stated domain shift requirement: local contrast normalisation through CLAHE and illumination-invariant structure detection through Black-Hat morphology. The result is a low-compute, interpretable baseline that can be deployed on new hardware without retraining.

The most direct path to improving centerline continuity would be to transition to a supervised deep learning approach. A Residual U-Net architecture trained on dense, manually annotated binary masks would likely resolve the fragmentation seen in low-contrast image regions. However, this would require collecting high-quality ground truth labels first, which the current dataset does not provide. Until that labeling effort is complete, the classical pipeline presented here represents the most robust option available.

8. References

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